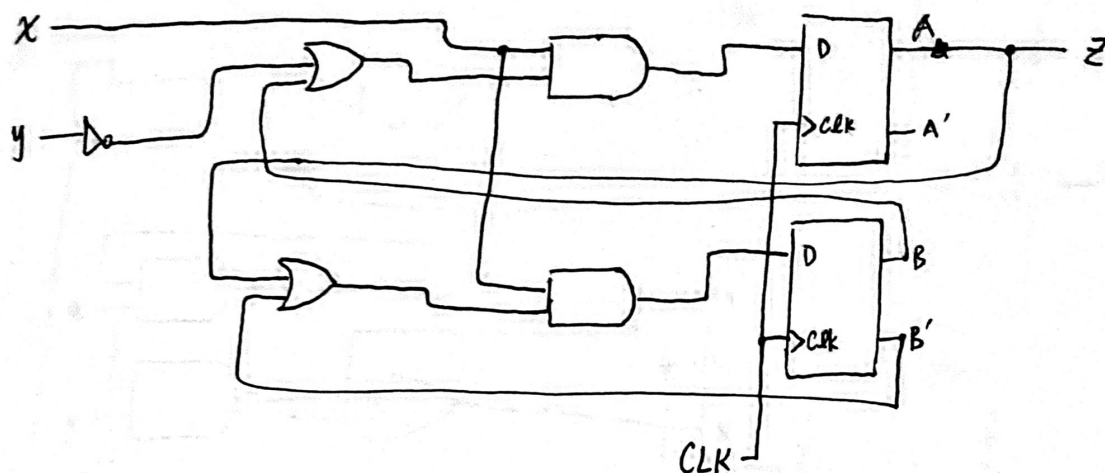


1. $A(t+1) = xy' + xB = x(y' + B)$
 $B(t+1) = xA + xB' = x(A + B')$
 $z = A$

a)



b)

~~現態 輸入 次態 輸出~~

~~A B x A B~~

現態 輸入 次態 輸出

A B x y A B z

~~xx 0~~

0 x 0 x 0 0 0

1 x 0 x 0 0 1

0 0 1 1 0 1 0

1 0 1 1 0 1 1

0 1 1 x 1 0 0

1 x 1 0 1 1 1

0 0 1 0 1 1 0

1 1 1 1 1 1 1

整理後

現態 輸入 次態 輸出

A B x y A B z

0 x 0 x 0 0 0

0 0 1 0 1 1 0

0 0 1 1 0 1 0

0 1 1 x 1 0 0

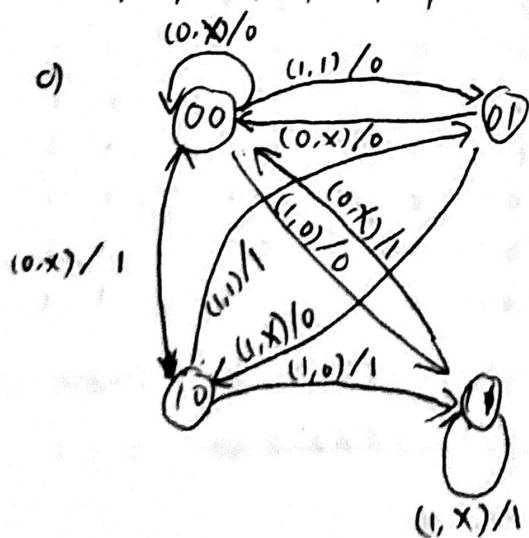
1 x 0 x 0 0 1

1 x 1 0 1 1 1

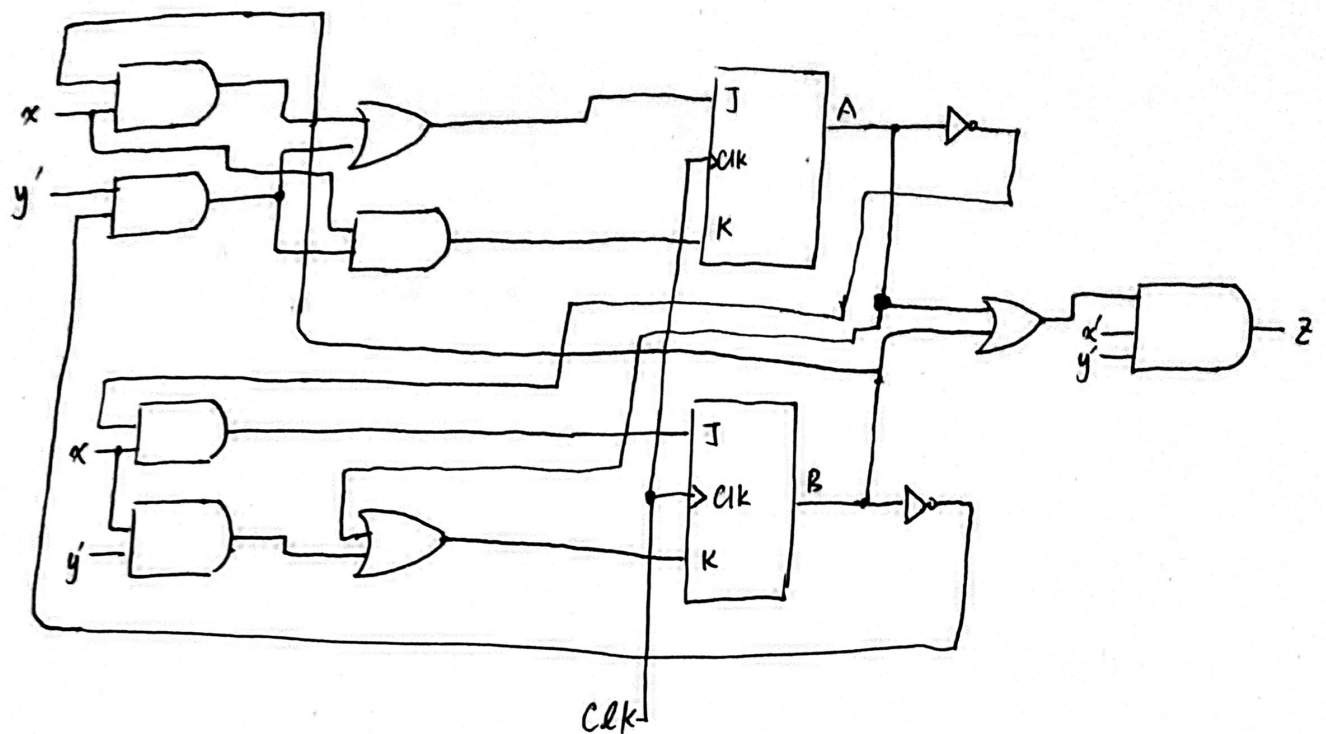
1 0 1 1 0 1 1

1 1 1 1 1 1 1

c)



a)



b)

現態		輸入		次態		輸出		觸發器輸入			
A	B	x	y	A	B	Z		J_A	K_A	J_B	K_B
0	0	0	0	1	0	0		1	0	0	0
0	0	0	1	0	0	0		0	0	0	0
0	0	1	0	1	1	0		1	1	1	1
0	0	1	1	0	1	0		0	0	1	0
0	1	0	0	0	1	1		0	0	0	0
0	1	0	1	0	1	0		0	0	0	0
0	1	1	0	1	0	0		1	0	1	1
0	1	1	1	1	1	0		1	0	1	0
1	0	0	0	1	0	1		1	0	0	1
1	0	0	1	1	0	0		0	0	0	1
1	0	1	0	0	0	0		1	1	0	1
1	0	1	1	0	0	0		0	0	0	1
1	1	0	0	0	0	1		0	0	0	1
1	1	0	1	0	0	0		0	0	0	1
1	1	1	0	1	0	0		1	0	0	1
1	1	1	1	1	0	0		1	0	0	1

$$c) A(z+1) = A'B'y' + A'Bx + AB'x' + ABx = B'(A'y' + Ax') + Bx$$

$$B(t+1) = \cancel{A'B'x + A'By} + A'[B'x + By]' = A'(B+x)(B'+x+y)$$

3. 傳播延時, $28 \mu s$

建立時間, $0.07 \mu s$